Worked examples — one route through each problem type

Cheat sheet · All questions · Engineering notes · R guide

Use the recognition cue to choose a method, then cover the solution and try the calculation. Examples use supplied course questions unless marked **adapted** or **toy**. Extra subparts show closely related skills without repeating whole questions. Numerical results were checked using R; retain full precision until the final answer. In $N(\mu, \sigma^2)$ the second parameter is **variance**.

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1. Event algebra and valid probabilities

Recognise: "A or B, given C" requires a union inside a conditional probability. **Source:** 2025 paper Q13–14.

Given $P(A) = 0.2$, $P(B) = 0.1$, $P(C) = 0.4$, $A \cap B = \varnothing$, B independent of C, and $P(A \cap C) = 0.1$:

$$P(B \cap C) = P(B)P(C) = 0.04,$$

$$P(A \cup B \mid C) = \frac{P(A \cap C) + P(B \cap C)}{P(C)} = \frac{0.10 + 0.04}{0.40} = 0.35.$$

There is no triple-intersection subtraction because A and B are disjoint. To check any proposed assignment, use

$$\max(0, P(A) + P(B) - 1) \le P(A \cap B) \le \min(P(A), P(B)).$$

For $P(A) = 0.5$, $P(B) = 0.7$, the intersection must be at least 0.2: a proposed 0.1 is invalid because it would make the union 1.1. The assignment $0.7, 0.2, 0.1$ is valid and gives union 0.8.

Trap: disjoint events with positive probabilities are not independent. Independence would require a positive intersection.

2. Total probability, Bayes and a probability tree

Recognise: an imperfect test gives conditional probabilities; the observed positive-test frequency can reveal prevalence. **Source:** Prac 1 Q1.

Let A = pipe has corrosion and B = test indicates corrosion. Given $P(B \mid A) = 0.7$, $P(B \mid A') = 0.2$, $P(B) = 0.25$, set $p = P(A)$.

$$0.25 = P(B \mid A)p + P(B \mid A')(1 - p) = 0.7p + 0.2(1 - p), \qquad p = 0.1.$$

The tree's terminal branches are:

First branch	Second branch	Multiply along the path
Corrosion A: 0.1	Positive B: 0.7	$P(A \cap B) = 0.07$
Corrosion A: 0.1	Negative B': 0.3	$P(A \cap B') = 0.03$
No corrosion A': 0.9	Positive B: 0.2	$P(A' \cap B) = 0.18$
No corrosion A': 0.9	Negative B': 0.8	$P(A' \cap B') = 0.72$

Add the positive branches: $P(B) = 0.07 + 0.18 = 0.25$. Reverse the condition by dividing the appropriate joint probability by the observed event:

$$P(A \mid B) = \frac{0.07}{0.25} = 0.28, \qquad P(A \mid B') = \frac{0.03}{0.75} = 0.04.$$

Trap: $P(B \mid A)$ is not $P(A \mid B)$. The identity to memorise is

$$P(B \mid A)P(A) = P(A \cap B) = P(A \mid B)P(B).$$

The multiplier on the right is $P(B)$.

3. Multiple causes and event-tree paths

Recognise: combine different scenarios, each with its own conditional outcome probability. **Source:** Lecture 1 Q1.

Treat the four stated causes A, B, C, D as mutually exclusive and exhaustive, with probabilities $0.30, 0.40, 0.15, 0.15$. Conditional explosion probabilities are $0.25, 0.20, 0.40, 0.75$.

$$P(E) = \sum_i P(i)P(E \mid i) = 0.075 + 0.080 + 0.060 + 0.1125 = 0.3275.$$

Given that an explosion occurred, the probability assigned to static electricity is

$$P(A \mid E) = \frac{0.075}{0.3275} = 0.2290.$$

Adapted event-tree extension: a leak occurs with probability 0.02 per mission; ignition given a leak has probability 0.3; suppression fails given an ignited leak with probability 0.1. The major-fire path is

$$P(\text{major fire}) = 0.02(0.3)(0.1) = 0.0006 \text{ per mission}.$$

Trap: multiply conditional probabilities along a path and add disjoint outcome paths. Do not sum overlapping possible causes as though they formed a partition. A frequency initiating an event tree produces a frequency at its endpoint; a probability produces a probability.

4. Normalise a PDF and obtain its CDF and moments

Recognise: the density contains an unknown constant. **Adapted from:** Stats Quiz 1 Q8, replacing its known coefficient by c.

Let $f(x) = \text{cx}$ for $0 < x < 8$, zero otherwise. Total area must be one:

$$1 = \int_0^8 \text{cx}\,\text{dx} = 32c \quad \Rightarrow \quad c = \frac{1}{32}.$$

Then

$$F(x) = \begin{cases} 0 & x \le 0 \\ \frac{x^2}{64} & 0 < x < 8 \\ 1 & x \ge 8. \end{cases}$$

$$P(1.6 < X < 4.1) = \frac{4.1^2 - 1.6^2}{64} = 0.22265625 \approx 0.22.$$

$$E[X] = \int_0^8 \frac{x^2}{32}\,\text{dx} = \frac{16}{3}, \qquad E[X^2] = \int_0^8 \frac{x^3}{32}\,\text{dx} = 32,$$

$$\text{Var}(X) = 32 - \left(\frac{16}{3}\right)^2 = \frac{32}{9}, \qquad \text{SD}(X) = \frac{\sqrt{32}}{3} \approx 1.8856.$$

For the median, $F(m) = 0.5$ gives $m = \sqrt{32} \approx 5.6569$.

Trap: a density height is not a point probability; $P(X = x) = 0$ for a continuous variable. Variance requires subtracting the squared mean.

5. Piecewise PDF and CDF to PDF

Recognise: the integration rule changes at a breakpoint. **Sources:** Lecture 1 Q2, Stats Quiz 1 Q7.

Repair time has $f(x) = \frac{x}{1600}$ for $0 < x \le 40$ and $\frac{80-x}{1600}$ for $40 < x < 80$, zero elsewhere. Integrating from the beginning of the support gives

$$F(x) = \begin{cases} 0 & x \le 0 \\ \frac{x^2}{3200} & 0 < x \le 40 \\ 1 - \frac{(80-x)^2}{3200} & 40 < x < 80 \\ 1 & x \ge 80. \end{cases}$$

The second branch includes the area already accumulated to 40. Therefore

$$P(60 \le X \le 70) = F(70) - F(60) = \frac{20^2 - 10^2}{3200} = 0.09375.$$

For an interval crossing the breakpoint (added subpart),

$$P(30 < X < 50) = F(50) - F(30) = 1 - \frac{30^2}{3200} - \frac{30^2}{3200} = 0.4375.$$

Symmetry gives $E[X] = 40$ h; equivalently integrate $xf(x)$ separately over $(0, 40)$ and $(40, 80)$.

Reverse direction: if $F(x) = \frac{(x+1)^2}{9}$ on $-1 < x \le 2$, with zero below and one above, then $f(x) = F'(x) = 2\frac{x+1}{9}$ on $(-1, 2)$ and zero elsewhere. Check the CDF is continuous at the boundaries; a jump would represent point mass.

Trap: do not start the second CDF branch at zero. Cumulative probability does not reset at a breakpoint.

6. Binomial counts and at least one

Recognise: fixed n, independent trials, two outcomes and constant p. **Source:** Prac 2 Q1.

For ten resistors with defect probability 0.08, $X \sim \text{Binomial}(10, 0.08)$:

$$P(X > 1) = 1 - P(X = 0) - P(X = 1) = 1 - 0.92^{10} - 10(0.08)(0.92)^9 = 0.1879.$$

Added subparts: $E[X] = \text{np} = 0.8$, $\text{SD}(X) = \sqrt{np(1-p)} = \sqrt{0.736} \approx 0.8579$. The probability of at least one defect is $1 - 0.92^{10} \approx 0.5656$.

```r
pbinom(1, size = 10, prob = 0.08, lower.tail = FALSE)
```

Trap: $P(X \geq k) = 1 - P(X \leq k-1)$. Discrete endpoints matter.

7. Poisson counts and independent arrival streams

Recognise: events counted over time at a specified rate. **Sources:** 2025 Q15, Prac 2 Q2.

Repair requests arrive at one per week. Over four weeks $N \sim \text{Poisson}(4)$, so

$$P(N = 3) = \frac{4^3 e^{-4}}{3!} = 0.195367 \approx 0.195.$$

The parameter is rate times interval, $\mu = \lambda t$, and both mean and variance equal 4.

For independent elevator/passenger streams at rates 1 and 5 per minute, respectively, the chance of no elevator and at least ten passengers in one minute is

$$e^{-1}\left[1 - \sum_{k=0}^{9} \frac{5^k e^{-5}}{k!}\right] = 0.011709.$$

```r
dpois(0, 1) * ppois(9, 5, lower.tail = FALSE)
```

Trap: the rate is not automatically the Poisson parameter. Independence of the two arrival streams is an assumption needed for the product.

8. Uniform probabilities and quantiles

Recognise: constant density over a finite interval. **Source:** Lecture 2, hose model $X \sim U(85, 165)$.

The PDF height is $\frac{1}{165-85} = \frac{1}{80}$. Thus

$$P(X < 120) = \frac{120 - 85}{80} = 0.4375.$$

For 99% survival to age c,

$$P(X > c) = \frac{165 - c}{80} = 0.99 \quad \Rightarrow \quad c = 85.8 \text{ h}.$$

Added summaries: $E[X] = 125$ h, $\text{SD}(X) = \frac{80}{\sqrt{12}} \approx 23.0940$ h. On $85 < t < 165$, $R(t) = \frac{165-t}{80}$ and $h(t) = \frac{1}{165-t}$: a flat density does not mean constant hazard.

Trap: clip probability intervals to the support; the CDF is zero below 85 and one above 165.

9. Normal tails, quantiles and unknown parameters

Recognise: standardise a boundary for a probability; invert a CDF for a boundary. **Sources:** Lecture 2, Stats Quiz 2 Q2.

For $X \sim N(125, 14^2)$,

$$P(X < 120) = \Phi\left(\frac{120 - 125}{14}\right) = 0.3605.$$

For $P(X > c) = 0.99$, only 0.01 lies below c:

$$c = 125 + 14\Phi^{-1}(0.01) = 92.43 \text{ h}.$$

For $P(|Z| > z) = 0.85$, each tail is 0.425, so $z = \Phi^{-1}(0.575) = 0.1891 \approx 0.19$.

Adapted parameter-finding subpart: if $X \sim N(\mu, 10^2)$ and $P(X < 400) = 0.05$, then $\frac{400-\mu}{10} = z_{0.05} = -1.64485$, giving $\mu = 416.4485$.

More generally two known quantiles x_1, x_2 at probabilities p_1, p_2 give

$$\sigma = \frac{x_2 - x_1}{z_{p_2} - z_{p_1}}, \qquad \mu = x_1 - \sigma z_{p_1}.$$

Trap: R's `sd` is the standard deviation. A normal lifetime model allows negative values; check whether that is negligible for the application.

10. Lognormal moments, tails and survival given age

Recognise: $\ln T$ is normal or $T = \exp(X)$ for normal X. **Sources:** 2025 Q16, Prac 3 Q5.

For a repair time $T = e^X$ with $X \sim N(1.2, 4.3)$,

$$P(T > 4) = P(X > \ln 4) = 1 - \Phi\left(\frac{\ln 4 - 1.2}{\sqrt{4.3}}\right) = 0.464207 \approx 0.464.$$

For the light-bulb problem $\ln T \sim N(10, 0.2^2)$ (log-scale parameters; T measured in hours):

$$E[T] = e^{10+\frac{0.2^2}{2}} = 22471.43 \text{ h},$$

$$\text{SD}(T) = e^{10+\frac{0.2^2}{2}}\sqrt{e^{0.2^2} - 1} = 4539.61 \text{ h}.$$

Also $E[\ln T] = 10$ and $\text{Var}(\ln T) = 0.04$. The probability of surviving to 30000 h is

$$R(30000) = 1 - \Phi\left(\frac{\ln 30000 - 10}{0.2}\right) = 0.061202.$$

Given survival to 20000 h, survival to a **total age** of 30000 h has probability

$$\frac{R(30000)}{R(20000)} = 0.089307.$$

Trap: "another 30000 h after 20000 h" would instead require $\frac{R(50000)}{R(20000)}$. The lognormal is not memoryless; e^{μ} is its median, not its mean.

11. Gamma tails, quantiles and expected cost

Recognise: a positive gamma variable, often evaluated by software. **Sources:** Stats Quiz 2 Q5–6, Prac 1 Q3.

With shape $\alpha = 2.27$ and rate $\lambda = 1.22$,

```
pgamma(2.48, shape = 2.27, rate = 1.22, lower.tail = FALSE)
# 0.2506520
```

Hence $P(X > 2.48) = 0.251$. For shape 2.26, rate 1.27, and $P(X \le c) = 0.78$:

```
qgamma(0.78, shape = 2.26, rate = 1.27)
# 2.519904
```

For the effluent model $X \sim \Gamma(2, \text{rate} = 1)$, integration by parts gives

$$f(x) = \text{xe}^{-x}, \qquad F(x) = \int_0^x \text{ue}^{-u}\,\text{du} = 1 - (1 + x)e^{-x}, \quad x > 0.$$

Mean and variance are both 2. A test exceeds threshold a with probability $(1 + a)e^{-a}$. If each exceedance costs C, two weekly tests at threshold 4 have expected fortnightly cost

$$2C(1 + 4)e^{-4} = 0.183156C,$$

whereas one fortnightly test at threshold 3 costs $4\text{Ce}^{-3} = 0.199148C$ on average. The first has lower expected cost under these assumptions.

Trap: `scale = 1/rate`. The expected number of fines can be fractional; linearity of expectation does not require independent tests.

12. Exponential parameter finding and memorylessness

Recognise: constant hazard or an exponential lifetime, with a survival probability supplied. **Source:** Week 5 Quiz Q3 added summaries and conditional-survival subparts.

Given $R(1000) = 0.46$,

$$0.46 = e^{-1000\lambda} \quad \Rightarrow \quad \lambda = -\frac{\ln 0.46}{1000} = 0.000776529 \text{ h}^{-1}.$$

$$P(T \le 336) = 1 - e^{-336\lambda} = 0.229653 \approx 0.230.$$

Mean life and standard deviation are both $\frac{1}{\lambda} = 1287.78$ h; median life is $\ln \frac{2}{\lambda} = 892.62$ h.

For an item already aged 1000 h,

$$P(T > 1336 \mid T > 1000) = \frac{e^{-1336\lambda}}{e^{-1000\lambda}} = e^{-336\lambda} = 0.770347.$$

Its chance of failing in the next 336 h is again 0.229653. By contrast, the unconditional chance of failing between ages 1000 and 1336 is $R(1000) - R(1336) = 0.46(0.229653) \approx 0.105640$.

Trap: an unconditional interval probability is a difference of survivals; a conditional probability divides by survival to the starting age. "Good as new after repair" alone does not establish exponential lifetimes.

13. Weibull lifetime calculations

Recognise: given shape β and characteristic life η. **Sources:** Prac 3 Q3, Lecture 3, 2025 Q8.

For solar cells with $\beta = 2.5$, $\eta = 10$ years:

$$R(t) = e^{-\left(\frac{t}{10}\right)^{\{2.5\}}}, \qquad h(t) = \frac{2.5}{10}\left(\frac{t}{10}\right)^{1.5}.$$

Age	Hazard, year$\{\}^{-1}$
5 yr	0.088388
10 yr	0.250000
20 yr	0.707107

The increasing hazard represents wear-out. The mean and standard deviation are

$$E[T] = 10\Gamma(1.4) = 8.87264 \text{ yr},$$

$$\text{SD}(T) = 10\sqrt{\Gamma(1.8) - \Gamma(1.4)^2} = 3.79667 \text{ yr}.$$

The supplied rounded gamma values yield slightly different final decimals. Use the precision requested by the question.

For the 2025 foundation model, $\beta = 1.2$, $\eta = 14$ yr:

$$F(39) = 1 - e^{-\left(\frac{39}{14}\right)^{\{1.2\}}} = 0.967261 \approx 0.967.$$

For any Weibull, the pth failure quantile is $t_p = \eta[-\ln(1-p)]^{\frac{1}{\beta}}$. The lecture's $\beta = 0.5, \eta = 125$ model gives median $125(\ln 2)^2 = 60.06$ h, mean $125\Gamma(3) = 250$ h, and decreasing hazard. In both models $F(\eta) = 1 - e^{-1} = 0.632121$.

Adapted residual-life check: for $\beta = 2, \eta = 5$, compare

$$P(T > 6 \mid T > 4) = e^{-\left[\left(\frac{6}{5}\right)^2 - \left(\frac{4}{5}\right)^2\right]} = e^{-0.8} = 0.449329$$

with $P(T > 2) = e^{-0.16} = 0.852144$. The Weibull is memoryless only at $\beta = 1$.

Trap: η is neither generally the mean nor the median. Hazard is a rate conditional on survival, not a failure probability.

14. Convert reliability to density, hazard and MTTF

Recognise: a function is supplied instead of a named distribution. **Source:** Stats Quiz 3 Q1, Q2, Q10.

With $R(t) = \frac{3.24}{[(t+2.7)(t+1.2)]}$, $t \ge 0$, differentiate:

$$f(t) = -R'(t) = \frac{3.24(2t + 3.9)}{(t + 2.7)^2(t + 1.2)^2}, \qquad f(2.2) = 0.096889 \approx 0.10.$$

For $R(t) = \frac{2.08}{[(t+1.6)(t+1.3)]}$, logarithmic differentiation is easier:

$$h(t) = -\frac{d}{\mathrm{dt}} \ln R(t) = \frac{1}{t+1.6} + \frac{1}{t+1.3}, \qquad h(2) = 0.580808 \approx 0.58.$$

For $R(t) = (1+t)^{-7}$,

$$\mathrm{MTTF} = \int_0^\infty R(t)\,\mathrm{dt} = \left[-\frac{(1+t)^{-6}}{6}\right]_0^\infty = \frac{1}{6} \approx 0.17.$$

Adapted reverse-direction check: if $h(t) = 2\frac{t}{9}$, then $H(t) = \int_0^t 2\frac{u}{9}\,\mathrm{du} = \frac{t^2}{9}$, $R(t) = e^{-\frac{t^2}{9}}$, $F(t) = 1 - R(t)$ and $f(t) = \left(2\frac{t}{9}\right)e^{-\frac{t^2}{9}}$. This is Weibull ($\beta = 2, \eta = 3$).

Trap: $f = \mathrm{hR}$, not $F = \mathrm{hR}$. Check $R(0) = 1$, non-increasing survival, and total density area one. The survival-integral MTTF formula assumes nonnegative lifetimes.

15. Series systems and minimum lifetime

Recognise: the system fails as soon as any required component fails. **Sources:** Week 5 Quiz Q4, Prac 2 Q7, Week 4 quiz.

For independent exponential component lifetimes with mean lives $15020, 1200, 10000, 11000$ h,

$$\lambda_{\mathrm{sys}} = \sum_i \lambda_i = \frac{1}{15020} + \frac{1}{1200} + \frac{1}{10000} + \frac{1}{11000},$$

$$R_{\mathrm{sys}}(500) = e^{-500\lambda_{\{\mathrm{sys}\}}} = 0.579604 \approx 0.580.$$

The same argument handles $Y = \min(X_1, X_2, X_3)$ with rates $0.002, 0.003, 0.005$: $Y \sim \mathrm{Exp}(0.010)$, $P(Y > 300) = e^{-3} = 0.049787$, and $E[Y] = 100$ h. The Prac 2 solution's printed 0.0005 in one product is inconsistent with its stated 0.005 and numerical result.

Units/checking variant: five sensors each at 1.91 failures per million hours give $\lambda_{\mathrm{module}} = 9.55 \times 10^{-6}$ h$\{\}^{-1}$. Using 365.25 days/year,

$$R(10\ \mathrm{yr}) = e^{-9.55\times 10^{\{-6\}}(10)(365.25)(24)} = 0.432941.$$

The local Week 4 quiz lists 0.426 as its selected answer, but that does not follow exactly from its stated 1.91 rate. The workshop data instead give about 1.95 per million h; using 1.95 and 365 days/year gives 0.425666, explaining the selected option. See the engineering notes for the source-data calculation. Its stated whole-system rate 15.44 per million h gives mean life $\frac{10^6}{15.44} = 64766.84$ h $= 7.3884$ yr, below the ten-year requirement.

Trap: multiply reliabilities, not component mean lives. Sum hazards for independent series components; summing constant rates yields an exponential system only when the component hazards are constant.

16. Parallel and mixed systems

Recognise: redundancy means the subsystem fails only if all alternative paths fail. **Sources:** 2025 Q4, Stats Quiz 4 Q1.

Three independent a components in parallel, then b and c in series, with $a = 0.62, b = 0.91, c = 0.70$:

$$R_a = 1 - (1-a)^3 = 1 - 0.38^3,$$

$$R_{\mathrm{sys}} = R_a\, b\, c = (1 - 0.38^3)(0.91)(0.70) = 0.602047 \approx 0.60.$$

For independent exponential A, B in parallel and C in series, rates $a = 0.14, b = 0.04, c = 0.23$:

$$R_{\mathrm{sys}}(t) = \left[e^{-\mathrm{at}} + e^{-\mathrm{bt}} - e^{-(a+b)t}\right]e^{-\mathrm{ct}}.$$

The parallel lifetime is $\max(T_A, T_B)$; the complete system lifetime is $\min(\max(T_A, T_B), T_C)$. Integrating survival gives

$$E\left[T_{\mathrm{sys}}\right] = \frac{1}{a+c} + \frac{1}{b+c} - \frac{1}{a+b+c} = 3.967382.$$

Trap: the final exponential term is subtracted. Adding all three gives $R(0) = 3$, impossible. A parallel combination of exponentials is generally not exponential, so $\frac{1}{h(t)}$ is not generally its MTTF.

17. Condition on a bridge component

Recognise: a cross-link prevents immediate series/parallel reduction. **Toy example for the conditional-network method in the engineering reliability material.**

Five independent links have reliability 0.9. Two paths connect source S to destination D: upper $S \to U \to D$ has links A, B; lower $S \to V \to D$ has C, D_1. A bidirectional bridge link E joins U and V.

When E fails, the two series paths act in parallel:

$$R_{\text{sys}\,|E'} = 1 - (1 - R_A R_B)\left(1 - R_C R_{D_1}\right) = 1 - \left(1 - 0.9^2\right)^2 = 0.9639.$$

When E works, U and V effectively merge. The input links A, C are parallel, followed in series by parallel output links B, D_1:

$$R_{\text{sys}\,|E} = \left[1 - (1 - 0.9)^2\right]^2 = 0.9801.$$

Total probability gives

$$R_{\text{sys}} = R_E R_{\text{sys}\,|E} + (1 - R_E) R_{\text{sys}\,|E'} = 0.9(0.9801) + 0.1(0.9639) = 0.97848.$$

Trap: paths sharing a component are not independent, even if components are independent. Condition on the shared component or bridge, simplify each state, then combine.

18. Complete-data parameter estimation and model choice

Recognise: observed times are data, not supplied population parameters. **Source:** Lecture 4, fabricated hose repair data in the Westralia teaching case.

For $x = (10, 17, 24, 30, 14, 18, 15, 13, 21)$ minutes, $n = 9$ and $\bar{x} = 18$.

Model	Maximum-likelihood estimates
Normal	$\hat{\mu} = 18$; $\hat{\sigma} = \sqrt{\frac{\sum (x_i - 18)^2}{9}} = 5.811865$
Lognormal	$\hat{\mu} = \text{mean}(\ln x) = 2.840222$; $\hat{\sigma} = \sqrt{\frac{\sum (\ln x_i - \hat{\mu})^2}{9}} = 0.315505$
Exponential	$\hat{\lambda} = \frac{n}{\sum} x_i = \frac{1}{18} = 0.055556$ min$\{\}^{-1}$

The ordinary sample standard deviation is $s = \sqrt{\frac{\sum (x_i - 18)^2}{8}} = 6.164414$ min. Report which estimator is requested; R's `sd(x)` uses $n - 1$.

For model checking, plot a density histogram and fitted density; check positivity, skewness and tail fit. Here $\frac{s}{\bar{x}} = 0.3425$ is far below the exponential model's population SD/mean of 1. That is a reason to question exponential fit, although nine observations cannot decisively identify the true distribution.

Simulation and graphical-summary subpart: Stats Quiz 1 Q5 asks for a reproducible sample of twenty $N(10, 5^2)$ observations. Generate once and retain the vector:

```
set.seed(5507)
x <- rnorm(20, mean = 10, sd = 5)
c(mean = mean(x), median = median(x), sd = sd(x))
# mean 11.366560; median 11.349722; sd 6.036354
hist(x, probability = TRUE)
curve(dnorm(x, mean = 10, sd = 5), add = TRUE, col = "red")
boxplot(x)
```

The requested median is 11.350. The realised sample mean and SD differ from the population values 10 and 5 because this is one finite random sample. A density histogram, rather than raw counts, makes the theoretical density overlay comparable. Resetting the seed reproduces the sample; repeatedly calling `rnorm` without resetting generates new data.

Trap: lognormal parameters are estimated from **logs of the observations**, not the log of the sample mean. A histogram suggests candidate models; it does not prove the hazard shape.

19. Total time on test and censored exponential estimation

Recognise: surviving units contribute time but no failure. **Source:** 2025 Q3 and Q10.

Seven systems start together. Four fail at $3, 13, 25, 40$ months; three survive to 48 months without replacement. Total exposure is

$$\mathcal{T} = 3 + 13 + 25 + 40 + 3(48) = 225 \text{ component-months.}$$

With four failures and an exponential model,

$$\hat{\lambda} = \frac{4}{\left(\frac{225}{12}\right)(365.25)(24)} = 24.33645 \text{ failures per million h.}$$

The fitted mean life is $\frac{\mathcal{T}}{4} = 56.25$ months. This is a model-based estimate incorporating censored exposure, not the average of the four observed failure ages.

Failed-test variant: 113 foundation tests include five unsuccessful equipment tests with no usable durations supplied. Excluding these leaves 108 valid foundations: five fail at $3, 5, 6, 8, 12$ months and 103 survive to 24 months.

$$\mathcal{T} = 3 + 5 + 6 + 8 + 12 + 103(24) = 2506 \text{ months.}$$

The answer follows the paper's exclusion convention. If usable observation durations for the unsuccessful tests were known, they would require a stated censoring/data-quality decision.

Trap: do not use 7×48 when failed units leave the test, and do not divide failures by calendar duration alone when multiple units are observed. $\sum \frac{t_i}{r}$ estimates MTTF under an exponential model; it is not a universal censored-data mean estimator.

20. Repairable systems and interfailure times

Recognise: timestamps describe repeated failures of one operating asset. **Source:** Stats Quiz 4 Q6–7.

Five failures occur at operating times $1, 2, 3, 5, 8$ and observation continues to 17.5. Total operating exposure is 17.5, so

$$\widehat{\text{MTBF}} = \frac{17.5}{5} = 3.5.$$

For the separate timestamps $2, 3, 6, 9, 10$, observed to 11, completed interfailure durations are

$$2, 3-2, 6-3, 9-6, 10-9 = 2, 1, 3, 3, 1.$$

Their mean is 2. The population-divisor descriptive SD used in the local answer is $\sqrt{\frac{4}{5}} = 0.894427$. The conventional sample SD, which R's `sd` returns, is $\sqrt{\frac{4}{4}} = 1$.

The final one-unit interval is right-censored, so it is not an observed complete duration. Under an exponential renewal model, all eleven exposure units give fitted mean $\frac{11}{5} = 2.2$ and fitted lifetime SD 2.2, a different quantity from the descriptive SD of completed intervals.

Trap: do not take the SD of absolute timestamps. Include final censored exposure in an exposure-based rate estimate. If times are calendar time with repair downtime, establish operating time before using it as exposure.

21. Kaplan–Meier with censoring and tied failures

Recognise: estimate survival without selecting a parametric lifetime distribution. **Toy dataset illustrating** Prac 4 Q3 the LMS `exa1.csv` is not supplied locally.

Five units have observations $(2, F), (3, C), (4, F), (4, F), (5, C)$, where F means failure and C censoring.

Time	At risk just before time	Failures	Censored	Survival after time
2	5	1	0	$1\left(1-\frac{1}{5}\right) = 0.8$
3	4	0	1	0.8
4	3	2	0	$0.8\left(1-\frac{2}{3}\right) = 0.266667$
5	1	0	1	0.266667

Thus $\hat{R}(3.5) = 0.8$, $\hat{R}(4) = 0.266667$ and $\hat{F}(4) = 0.733333$. Censoring changes the next risk set but does not itself cause a drop.

```
library(survival)
d <- data.frame(time = c(2,3,4,4,5), status = c(1,0,1,1,0))
fit <- survfit(Surv(time, status) ~ 1, data = d)
summary(fit, times = c(3.5, 4))
```

Trap: using $1 - \frac{3}{5} = 0.4$ ignores censoring before the final failures. A positive KM tail beyond the last observation does not identify the full mean lifetime. Standard KM relies on non-informative censoring; recorded suspensions are not automatically harmless.

22. Ranks and Weibull probability plots

Recognise: the question explicitly asks for a plotting position or parameters from a straight line. **Sources:** Stats Quiz 4 Q3, Prac 3 Q4.

Eleven units are tested until the ninth failure at 13.48. The requested **mean rank** is

$$\hat{F}(t_9) = \frac{9}{11+1} = 0.75.$$

For comparison, the empirical/KM fraction with no earlier censoring would be $\frac{9}{11} = 0.818182$, while the approximate median rank is $\frac{9-0.3}{11+0.4} = 0.763158$. These are different estimators, so use the named method.

For Weibull data, define $x = \ln t$ and $y = \ln[-\ln(1 - F(t))]$. Then

$$y = \beta x - \beta \ln \eta.$$

Toy fitted line: $y = 2x - 3.218876$. Reading slope and intercept gives $\hat{\beta} = 2$ and

$$\hat{\eta} = e^{-\frac{-3.218876}{2}} \approx 5.$$

Straightness supports a two-parameter Weibull fit; curvature can indicate a shift, mixture or poor model choice. In the supplied mixed plot in Week 5 Quiz Q6, the earlier segment has $\beta_1 = 1.5278 > \beta_2 = 1.3554$.

Adjusted median-rank subpart with suspensions: a toy dataset following the method in the Weibull slides is $(2, F), (3, C), (4, F), (5, F), (6, C)$. Here $n = 5$. Start with adjusted rank $j_{\text{prev}} = 0$; for each failure let m count all preceding observations and calculate

$$\text{INC} = \frac{n + 1 - j_{\text{prev}}}{1 + n - m}, \qquad j = j_{\text{prev}} + \text{INC}, \qquad \hat{F} = \frac{j - 0.3}{n + 0.4}.$$

Failure time	m	Increment	Adjusted rank j	Median-rank estimate
2	0	$\frac{6-0}{6} = 1$	1	$\frac{0.7}{5.4} = 0.129630$
4	2	$\frac{6-1}{4} = 1.25$	2.25	$\frac{1.95}{5.4} = 0.361111$
5	3	$\frac{6-2.25}{3} = 1.25$	3.50	$\frac{3.2}{5.4} = 0.592593$

The suspension at time 3 receives no failure point but increases m for the next update. Continue from the previous **adjusted** rank rather than resetting to the observed failure number. These are plotting positions, distinct from Kaplan–Meier estimates.

Trap: check which variable is on each axis; swapping axes changes the slope. Do not transform $F = 0$ or $F = 1$ because the logs diverge. With interspersed censoring use a censoring-aware CDF estimator, not unadjusted failure ranks.

23. Weibull maximum-likelihood estimation

Recognise: asked to fit shape and scale numerically. **Source:** Stats Quiz 4 Q10, which explicitly instructs omission of censored values.

Use these twenty observed failures:

```
x <- c(8,21.3,25.5,8.1,4.5,4.2,13.2,9.9,27,25.7,
       25.8,28.6,13.5,7,20.9,9.6,33.9,16.5,4.4,19.2)
score <- function(beta) {
```

```
  1/beta + mean(log(x)) - sum(x^beta * log(x))/sum(x^beta)
}
beta_hat <- uniroot(score, c(0.1, 10), tol = 1e-12)$root
eta_hat <- mean(x^beta_hat)^(1/beta_hat)
c(shape = beta_hat, scale = eta_hat)
# shape 1.888809; scale 18.459323
```

The score equation is

$$\frac{1}{\beta}+\frac{1}{n}\sum \ln x_i-\frac{\sum x_i^{\beta}\ln x_i}{\sum x_i^{\beta}}=0,$$

with $\hat{\eta}=\left[\sum \frac{x_i^{\hat{\beta}}}{n}\right]^{\frac{1}{\hat{\beta}}}$. Therefore the requested shape is 1.889.

If censored observations are to be retained, use their survival contribution. For observed times t_i and failure indicators δ_i:

$$\ell(\beta,\eta)=\sum_i \delta_i[\ln\beta-\beta\ln\eta+(\beta-1)\ln t_i]-\sum_{i\left(\frac{t_i}{\eta}\right)}^{\beta}.$$

Trap: omitting suspensions is required for this particular quiz calculation, but is generally a biased way to fit censored lifetime data. Never relabel a suspension as a failure. See the R guide for the censoring-aware fit.

24. Normal sample mean, sum and standard error

Recognise: the random variable is an average of n measurements. **Source:** 2025 Q17 sum subpart added from the Prac 5 method.

Loaded bucket volumes are iid $N(11.5,1.4^2)$ m{}^{3}. For $n=12$:

$$\bar{X}\sim N\left(11.5,\frac{1.4^2}{12}\right),\qquad \mathrm{SE}(\bar{X})=\frac{1.4}{\sqrt{12}}.$$

$$P(\bar{X}<10.88)=\Phi\left(\frac{10.88-11.5}{\frac{1.4}{\sqrt{12}}}\right)=0.062502\approx 0.063.$$

For total volume $S=\sum X_i$, $S\sim N(138,23.52)$, and $P(S<130.56)$ is the same probability because $130.56=12(10.88)$.

Trap: the sample mean uses SD $\frac{\sigma}{\sqrt{n}}$; the sum uses SD $\sigma\sqrt{n}$. Both results are exact for independent normal observations, even when $n<30$.

25. Minimum sample size

Recognise: find n to meet a probability or precision requirement. **Sources:** Stats Quiz 5 Q4, Lecture 5.

Normal observations have variance 16, so $\sigma=4$. Require $P(|\bar{X}-\mu|<2)\geq 0.99$. The middle 99% leaves 0.005 in each tail:

$$\frac{2}{\frac{4}{\sqrt{n}}}\geq z_{0.995}=2.575829,\qquad n\geq\left(\frac{4z_{0.995}}{2}\right)^2=26.539586.$$

Hence $n_{\min}=27$. Check: $n=26$ gives 0.989213, below the requirement; $n=27$ gives 0.990625, above it.

For the lecture's **one-sided** requirement with mean 300, SD 15, and $P(\bar{T}\geq 310)\leq 0.005$:

$$n\geq\left(\frac{15z_{0.995}}{310-300}\right)^2=14.928517,\qquad n_{\min}=15.$$

Trap: the first problem divides its total tail probability by two; the second already specifies one tail. Always round the required sample size up.

26. Student's t and two-sided critical values

Recognise: unknown population SD is estimated by S, or a t distribution is explicitly given. **Sources:** 2025 Q18, Stats Quiz 5 Q5.

For twenty independent normal lifetimes with unknown mean and variance,

$$\frac{\bar{X}-\mu}{\frac{S}{\sqrt{20}}} \sim t_{19} \text{ exactly.}$$

For $T \sim t_5$, find positive c satisfying $P(|T| > c) = 0.0125$. Each tail has area 0.00625, so

$$c = t_{5,0.99375} = 3.810005 \approx 3.810.$$

```
qt(1 - 0.0125/2, df = 5)
2 * pt(3.810004700, df = 5, lower.tail = FALSE) # about 0.0125
```

Trap: normality of the population makes the t result exact. A large sample from an unspecified distribution supports approximate normal inference, not an exact t law. When σ is known, the normal-population standardised mean uses Z, not t.

27. CLT for Weibull lifetimes

Recognise: the individual distribution is non-normal, but the question concerns a mean of many independent observations. **Source:** Prac 5 Q5.

For $T_i \sim \text{Weibull}(2, 5)$ and $n = 50$, first calculate the individual moments:

$$\mu = 5\Gamma(1.5) = 4.431135, \quad \sigma^2 = 25[\Gamma(2) - \Gamma(1.5)^2] = 5.365046.$$

For **one** board, $P(T_1 > 5) = e^{-1} = 0.367879$. For the **sample average**, the CLT gives

$$\bar{T} \dot{\sim} N\left(4.431135, \frac{5.365046}{50}\right),$$

$$P(\bar{T} > 5) \approx 1 - \Phi\left(\frac{5 - 4.431135}{\sqrt{\frac{5.365046}{50}}}\right) = 0.041226.$$

For $P(|\bar{T} - \mu| < 0.5) > 0.95$, the CLT sample-size calculation is

$$n > \left(\frac{z_{0.975}\sqrt{5.365046}}{0.5}\right)^2 = 82.438412, \qquad n_{\min} \approx 83.$$

Trap: a mean of Weibull observations is not itself generally Weibull. This calculation is approximate; a "30 observations" rule does not guarantee accurate tail probabilities. Normal populations give exact normal sample means; here the CLT is doing the work.

28. Shifted Weibull and age-replacement cost

Recognise: replace at failure or scheduled age, whichever happens first. **Source:** Week 5 Quiz Q5.

A cable has shape $\beta = 1.8$, scale $\eta = 500$ h and threshold $\gamma = 50$ h. Corrective replacement costs \$6000, scheduled replacement \$600; operation is $H = 5000$ h/year. Candidate ages are $50, 75, \ldots, 300$ h.

$$R(t) = \begin{cases} 1 & t < 50 \\ e^{-\left(\frac{t-50}{500}\right)^{\{1.8\}}} & t \geq 50. \end{cases}$$

Each replacement restores the cable to new condition. Ignoring downtime, the long-run annual cost is

$$C(m) = 5000\frac{6000[1 - R(m)] + 600R(m)}{\int_0^m R(t)\,\mathrm{dt}}.$$

At $m = 175$ h, $F(m) = 0.07916024$ and expected cycle length is 171.40895 h. Therefore

$$C(175) = 5000\frac{6000(0.07916024) + 600(0.92083976)}{171.40895} \approx \$29{,}971.17 \text{ per year.}$$

Candidate age (h)	Annual cost, rounded to nearest dollar
50	60,000
75	41,657
100	34,342
125	31,205
150	30,053
175	**29,971**
200	30,510
225	31,425
250	32,571
275	33,860
300	35,231

Thus 175 h is the minimum **among the specified candidates**. Compute the whole grid, not only neighbouring values. For accurate numerical integration, split at the threshold 50 or use tight tolerances.

Trap: dividing by scheduled age m ignores cycles ending early at failure. A fixed calendar/block replacement policy has a different cost model. Do not claim this grid optimum is the unconstrained continuous optimum.

29. Engineering calculations and short-answer patterns

Recognise: the task asks for an engineering decision, financial metric or classification. **Sources:** 2025 Q1, Q5–7, Q9, Q12 see engineering notes for full slide coverage and further case examples.

Discounted maintenance costs: four end-of-year payments of \$150000 at an 8% real discount rate give

$$\text{NPC} = 150000\sum_{t=1}^{4}(1.08)^{-t} = 496819.03, \qquad \$496{,}819.$$

A time-zero investment would be added without discounting. Positive costs are summed for NPC; NPV requires consistent signs for cash inflows and outflows.

Toy availability calculation: an asset averages 100 operating hours between failures and 5 repair hours per failure. Its long-run availability is

$$A = \frac{100}{100 + 5} = 0.952381.$$

Availability allows repair and return to operation; mission reliability is the probability of no failure over a specified duration.

Short-answer examples:

Problem type	Worked reasoning → answer	Trap
Classify a control	Sealing nitrocellulose packaging physically retains its damping agent → **engineering control**	A written instruction to check the seal would be administrative
Failure vocabulary	"Engine fails to start" describes lost function → **failure mode**; a flat battery can be a cause	Mode, cause and mechanism answer different questions
Prioritise FMEA	Cable connections have $S = 6, O = 7, D = 4$: RPN $= 168$, exceeding leakage's $7(6)(3) = 126$ → select cable connections in this question	RPN is an ordinal prioritisation tool, not failure probability or expected monetary loss
Select maintenance	Vibration/infrared evidence of deterioration → **condition-based maintenance**	The measuring technique supports the tactic; it does not itself define a fixed replacement age
Rate a scenario	In the supplied paper's matrix, event heard of in industry → likelihood B; multiple fatalities → consequence 5; answer **B5**	Use the provided matrix definitions, not invented numerical probabilities

Problem type	Worked reasoning → answer	Trap
Hidden protection failure	**Toy:** a standby trip can fail undetected until demanded; a feasible proof test reveals loss of protection → **failure-finding task**	A hidden failed device may show no developing P–F warning
Explain a P–F interval	**Toy:** detectable deterioration at day 10, functional failure at day 30 → interval 20 days; inspect sufficiently often to detect and act within it	Inspection spacing alone is insufficient if repair preparation takes longer than the remaining warning time

Engineering answers should connect the observed fact to the definition, then state the decision. Where a decision uses failure probability, hazard, reliability, availability, expected cost or uncertainty, the statistical calculations in this document provide the quantitative input.